4.10.12

EE25BTECH11011 - Navya Banavathu

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Question

The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

Solution

The equations of the given lines are

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \mathbf{x} = 10 \tag{1}$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \mathbf{x} = -5 \tag{2}$$

Form a single matrix equation (all three lines):

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 5 & 4 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 10 \\ -5 \\ 0 \end{pmatrix}. \tag{3}$$

Solution

To solve, take the first two rows and solve by inverse:

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 10 \\ -5 \end{pmatrix} \tag{4}$$

$$\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 10 \\ -5 \end{pmatrix} = \frac{1}{-3} \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 10 \\ -5 \end{pmatrix} = \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} \tag{5}$$

Solution

Now verify that this vector satisfies the third row of (3),

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \mathbf{x} = 0 \tag{6}$$

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} = \frac{1}{3}(-100 + 100) = 0 \tag{7}$$

Hence, The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

Code (C)

Part 1

```c
#include <stdio.h>

int find_intersection(double *x, double *y) {
    double a1 = 1, b1 = 2, c1 = 10;
    double a2 = 2, b2 = 1, c2 = -5;

    double det = a1*b2 - a2*b1;
    if(det == 0) return 0; // parallel

    *x = (c1*b2 - c2*b1) / det;
    *y = (a1*c2 - a2*c1) / det;

    if(5*(*x) + 4*(*y) == 0)
        return 1; // passes
    else
        return 2; // does not pass
}
```

Python Code:Import and Define points

Part 1

```
import ctypes
import matplotlib.pyplot as plt
import numpy as np

# --- Load the DLL ---
lib = ctypes.CDLL(r"C:\Users\Navya\code8.dll")

# Prepare variables to receive x and y from C
x = ctypes.c_double()
y = ctypes.c_double()

# Call the C function
result = lib.find_intersection(ctypes.byref(x), ctypes.byref(y))
xi, yi = x.value, y.value
```

Python Code

Part 2

```
print(f"Intersection point: ({xi:.2f}, {yi:.2f})")

if result == 0:
    print("Lines are parallel, no intersection.")
elif result == 1:
    print("The line 5x + 4y = 0 passes through the intersection
        point.")
else:
    print("The line 5x + 4y = 0 does NOT pass through the
        intersection point.")
```

Python Code: Plotting

Part 3

```python
# Plot the lines
x_vals = np.linspace(-10, 10, 400)

# Line 1: x + 2y = 10 => y = (10 - x)/2
y1 = (10 - x_vals)/2
# Line 2: 2x + y = -5 => y = -5 - 2*x
y2 = -5 - 2*x_vals
# Line 3: 5x + 4y = 0 => y = -5/4 * x
y3 = -5/4 * x_vals

plt.figure(figsize=(8,6))
plt.plot(x_vals, y1,label=r'$(1\ 2)\mathbf{x} = 10$',color='blue')
plt.plot(x_vals, y2,label=r'$(2\ 1)\mathbf{x} = -5$',color='green')
plt.plot(x_vals, y3,label=r'$(5\ 4)\mathbf{x} = 0$',color='red')
```

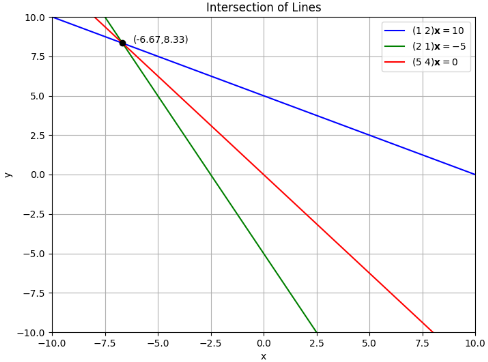
Python Code: Plotting

Part 4

```
# Mark the intersection point
plt.scatter(xi, yi, color='black', zorder=5)
plt.text(xi+0.5, yi, f'({xi:.2f},{yi:.2f})', fontsize=10)

plt.xlabel('x')
plt.ylabel('y')
plt.title('Intersection of Lines')
plt.grid(True)
plt.legend()
plt.xlim(-10, 10)
plt.ylim(-10, 10)

# Save figure as fig8.png
plt.savefig('fig8.png', dpi=300)
plt.show()
```


Intersection of Lines
(1 2)x = 10
(2 1)x = −5
(5 4)x = 0
(-6.67,8.33)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x
y